Research Plan — Technical Report

Taxing liquidity on a decentralized prediction market: who pays the fee and who captures the maker subsidy

Draft technical report laying the foundations for one or more Q1 submissions in DeFi / Information Systems venues (e.g., *Information Systems Research*, *Journal of Management Information Systems*, *Decision Support Systems*, *Electronic Commerce Research and Applications*; finance-crossover: *Journal of Financial Markets*, *Management Science*).

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1. Introduction

1.1 State of the art

Polymarket is the largest decentralized prediction market — an on-chain venue where users buy and sell binary outcome shares that settle to $1 (if the event occurs) or $0 (otherwise), so a share's price is a market-implied probability. Trading runs on a central limit order book (CLOB): a *maker* posts a resting limit order that provides liquidity, and a *taker* crosses the spread with a marketable order that consumes it. Settlement, custody, and order matching are recorded on the Polygon blockchain, so — unlike centralized exchanges — the near-complete trade history is publicly observable.

A cluster of 2025–2026 studies has established the market's first-order facts:

- **Who wins and loses.** Akey et al. (2026, SSRN 6443103), using the reconciled full history (~2.4M users, ~$67B volume, 2022–2026), find that a small minority captures nearly all gains (top 1% of positive-PnL users ≈ 76.5% of profits; ≈69% of users end negative) and that liquidity provision (posting limit/maker orders) is the single strongest predictor of finishing profitable (≈ +9 percentage points per standard-deviation increase in maker-volume share). Their sample ends on 29 March 2026, the day before Polymarket introduced fees on all markets, deliberately stopping there because the fee "materially changes the economics of trading" — so their account is the definitive statement of the *pre-fee* distribution and leaves the fee-and-rebate era untouched.
- **Skill vs. luck.** Gómez-Cram et al. (2026, SSRN 6617059), on ~1.72M accounts, use a sign-randomization test (repeatedly shuffle each account's buy/sell directions and compare actual profit to the resulting null distribution) and conclude that ~3% of accounts are persistently skilled, that market-makers plus that skilled minority (<3.5% of accounts) capture

> 30% of all gains, and that "the majority does not produce accuracy; it funds it."

- **Fill-side behavior.** Nechepurenko (2026, arXiv 2605.11640) attempted a behavioral tiering of *maker-side* participants on the on-chain fill tape but, in the current version, retracts the tiers — retaining only a one-cluster null and concluding that public fill records cannot identify market-making without the order/quote lifecycle (a ~2.3-day sample; no profit, no account-type layer).
- **Informed / insider trading.** Mitts & Ofir (2026, SSRN 6426778) build a per-(wallet, market) suspicion score for informed trading, using *newly created wallets* as one signal, and flag a large set of suspicious positions.
- **Foundations.** The Polymarket-v1 public archive (Qin & Yang, 2026, arXiv 2606.04217) supplies a reconciled, ground-truth trade record; Rahman et al. (2025, arXiv 2510.15612) give a systematization of knowledge (SoK) for decentralized-prediction-market microstructure and enumerate open research gaps (insider timing, coordination, MEV, replay).

1.2 The gap

The literature above answers *who profits* and *whether skill exists* — and it answers them for the *pre-fee* market, because the definitive who-wins study stops on the eve of the fee. It does not address the economic event that most directly reshapes those outcomes: Polymarket's introduction of a trading fee. Two questions about that event are untouched by all of the work in §1.1.

1. **Who bears the fee, and is it regressive.** No study measures the fee's *incidence* — who bears it, as a share of the value they trade — or whether it is *regressive*, falling proportionally more on small orders. The financial-economics literature has strong analogues — Umlauf (1993) and Colliard & Hoffmann (2017) on securities-transaction taxes, and the SEC tick-size pilot — but none on a prediction market, and only Whelan (working paper) models prediction-market fees theoretically.
2. **Who captures the maker subsidy, and who structurally are they.** Polymarket's fee is levied on takers and partly rebated to makers, so its ultimate beneficiaries are whichever makers keep that rebate net of trading losses. The concurrent papers show *makers win on average* but not which makers capture the rebate, nor the mechanism. Adverse selection — the tendency for a passive liquidity provider to be "picked off" by better-informed counterparties (Glosten & Milgrom, 1985) — is the natural candidate mechanism and has never been measured on this venue. Nor does any prior work resolve the beneficiary's on-chain account type — an externally owned account versus a smart-contract wallet, and among the latter which proxy implementation — which puts a concrete structural identity on who ends up capturing the transfer.

A third question — how participants change their behavior after the fee, in order size, entry, and exit — is of secondary interest here. It is theoretically ambiguous for a cost this small (a fee can both deter marginal flow and prune noise; Cipriani & Guarino, 2008), the market-quality side of it is partly addressed by concurrent microstructure work, and the staggered rollout affords too few treated categories for a well-powered causal estimate. We therefore examine it descriptively, as market context for the two incidence-and-capture questions above, rather than as a headline claim.

1.3 Why this matters for Polymarket (and DePMs broadly)

A prediction market is only socially useful if its prices aggregate information — which requires participation, and especially participation by the small, price-taking "crowd" whose flow the informed minority trades against. A fee that is regressive and that transfers value from the retail crowd to a professional market-making elite could erode exactly the participation that makes prices informative, while concentrating rents. Whether that is happening — and who ends up bearing versus capturing the fee — is a first-order market-design and welfare question for Polymarket specifically and for the design of decentralized prediction markets generally. Because the entire ledger is on-chain, it is also a rare setting where these distributional questions can be answered on the *near-complete population* rather than a sample.

1.4 Research questions

We pose one main question, decomposed into two sub-questions.

Main RQ — How to measure, from public on-chain data, the incidence and distributional consequences of introducing a transaction fee on a decentralized prediction market?

Each sub-question isolates one facet of that measurement. For each, we state the question, why it is worth answering, and the gap in the literature it closes.

Sub-RQ 1 — What is the incidence of the fee across order sizes and trader roles, and is it regressive? *Why, and the gap it closes.* Incidence — who actually bears a levy, as a share of the value they trade — is the first-order welfare question about any transaction cost, and it has never been measured for a prediction market. The equity-market transaction-tax literature (Umlauf, 1993; Colliard & Hoffmann, 2017) measures aggregate liquidity and volume effects but not per-trader incidence, because it lacks the trader-level, whole-population record that an on-chain venue provides. This answer establishes whether the fee is flat or regressive — the fact the rest builds on.

Sub-RQ 2 — Who captures the maker subsidy the fee funds, through what mechanism, and what is their structural identity? Specifically, what distinguishes the liquidity providers who keep the rebate from those who lose it, what role does adverse selection play, and what on-chain account type and behavioral actor do the beneficiaries correspond to? *Why, and the gap it closes.* The fee is levied on takers and partly rebated to makers, so its ultimate beneficiaries are whichever makers keep that rebate net of trading losses. The concurrent full-population studies (Akey et al., 2026; Gómez-Cram et al., 2026) establish the *pre-fee* picture — makers win *on average*, and liquidity provision is the strongest predictor of profit — but not *which* makers keep the fee-funded rebate, nor the mechanism, nor who structurally they are. Adverse selection — a passive maker being "picked off" by better-informed counterparties (Glosten & Milgrom, 1985) — is the classic candidate mechanism and has never been measured on this venue. And no prior work resolves the beneficiary's on-chain account type — an externally owned account versus a smart-contract wallet, and among the latter which proxy implementation — nor crosses that structural layer with a behavioral classification and profitability. This answer names the beneficiary of the transfer, explains why, and attaches a concrete structural identity to it.

Together, the two answers compose the main question. Sub-RQ 1 fixes *what* is charged and *on whom* — the incidence. Sub-RQ 2 follows the money to *who ultimately keeps it*, *why*, and *who they are* — the mechanism and the distributional consequence, down to a structural identity. Incidence plus capture-with-identity is precisely what "the incidence and distributional consequences of introducing the fee, measured from public on-chain data" means, so the two sub-answers are jointly sufficient for the main RQ. The market's behavioral response to the fee — whether order size and entry fall — is examined as context in §5.2, not as a third claim, for the reasons given in §1.2.

1.5 How we answer each (preview)

- Sub-RQ 1. Fee incidence is computed per taker order and per order-size decile in the mature-fee regime, with the staggered per-category rollout corroborating the onset. Result: the fee is regressive — the smallest orders bear roughly two to three times the rate of the largest (§5.1).
- Sub-RQ 2. Maker-side realised PnL is decomposed against a counterparty-informedness proxy, and the beneficiaries are then given a structural identity: every active wallet is classified by on-chain account type (from bytecode and storage) and by behavioral function (from activity timing) and crossed with capture. Result: most makers lose, and a professional, largely automated elite that escapes adverse selection captures the rebate; account type resolves to an onboarding channel, not a skill axis (§5.3–§5.4). The economic direction replicates Akey and Gómez-Cram; the adverse-selection mechanism and the account-type identity are new to this literature.

The market's behavioral response to the fee is reported as context, not as a research claim: a balanced event study holding category composition constant around each category's own fee-start measures order size and new-wallet inflow, netting out pre-trends, and finds no participation or volume response attributable to the fee — order size follows a pre-existing downtrend and new entry does not fall at onset; the earlier "orders halve / entry collapses" figures were panel-composition artifacts (§5.2).

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2. Methodology

We describe methods in plain terms; technical terms are explained at first use.

2.1 Realised profit-and-loss (PnL). For each fill we compute the trader's realised PnL from the resolved outcome: buying a share of the winning outcome at price p for s shares yields $(1 - p)\cdot s$; buying the loser yields $-p\cdot s$ (symmetrically for sells). Summing over a wallet's fills gives its realised PnL. This is *outcome-dependent* — it uses the eventual result, so it measures realised (not risk-adjusted or forward-looking) performance and is subject to survivorship (wallets that stopped trading are still counted only where they appear). We report it as descriptive, never as certified skill.

2.2 Maker vs. taker and the fee. On each on-chain `OrderFilled` event we identify the *maker* (resting-order owner) and *taker* (aggressor). The fee (in the platform's 6-decimal USD unit) is charged once per taker order and recorded on the taker-order record; we therefore read fee incidence from those records and PnL from the maker-fill records.

2.3 The fee rollout as a natural experiment. Fees were introduced staggered by market category over early 2026. We *recover* each category's fee-start month from the data (the first month its effective fee rate jumps above a threshold) and run an event study — outcomes aligned to "months relative to that category's own fee-start," with not-yet-treated categories serving as the implicit control (a difference-in-differences, DiD, design: comparing the change in treated units to the change in not-yet-treated units).

2.4 Fee incidence and regressivity. Incidence = fee paid ÷ notional traded, reported per order-size decile. "Regressive" means the rate is *higher* for smaller orders.

2.5 Adverse selection. A passive maker is *adversely selected* when the price moves against it right after it fills — informed flow picking it off. We measure this with an identified signed post-fill *markout*: for each maker fill, the token's forward price minus the fill price, signed by the maker's side, so a negative markout is adverse selection (Glosten & Harris, 1988; Dubach, 2026). We report it over horizons from 60 s to one day and by maker-volume decile.

2.6 Skill test (supporting). To separate skill from luck we use a sign-randomization (permutation) test: holding each wallet's positions, prices and outcomes fixed, we randomly flip the buy/sell direction of each *market position* many times; if the wallet's actual PnL is not better than this null, its side-selection is indistinguishable from chance. We report the fraction "certifiably skilled" both uncorrected and after Benjamini–Hochberg false-discovery-rate control (which bounds the expected share of false positives among the wallets we call skilled), and we cross-check with a correlation-immune split-half persistence test (does period-1 PnL predict period-2 PnL). We flag that this test's independence assumption is inflated by cross-market outcome correlation, so we treat the skilled fraction as an upper bound and the persistence result as the robust signal.

2.7 On-chain account-type classification. For a wallet, we call the Polygon node's `eth_getCode` to distinguish an EOA (externally owned account — no contract code, i.e. a raw key) from a smart-contract wallet. Contract wallets are keyed by their code hash and, for proxies, their implementation: an EIP-1167 minimal proxy embeds its implementation address in the bytecode; a Gnosis Safe proxy stores its singleton in a storage slot. This maps to Polymarket's documented `signature_type` (0 = EOA, 1 = Polymarket factory proxy for email/"Magic" users, 2 = Gnosis Safe for browser-wallet users). Classification is computed on an activity- and PnL-stratified sample plus the full set of top winners, with rate-limit-resilient, cached calls.

2.8 Behavioral-actor classification. Independently of the account type, we classify each active wallet (≥20 fills) by *function* using activity features: maker-share (fraction of fills as maker), fills per active day (intensity), and hour-of-day entropy (how evenly its trades spread across the 24 hours — a high value indicates near-24/7 activity consistent with automation). Heuristic labels: *market-maker* (maker-share ≥ 0.7), *hft_bot* (≥40 fills/active-day and near-24/7), *active_trader*, *retail_casual*, *one_shot*. Caveat (per Nechepurenko, 2026): public fills without the quote lifecycle cannot *prove* market-making; we therefore present these as heuristic behavioral classes, not identified market-maker status.

2.9 Entity / coordination (supporting, negative). We attempt to collapse wallets to beneficial owners via a shared-funding graph (wallets funded by a common non-hub USDC sender), excluding high-degree "hub" funders (exchange/relayer/platform contracts), and we build a co-trading graph (same market, same side, same day). These test for sybil concentration and coordinated syndicates.

2.10 Memory-safe, reproducible computation. All population-scale aggregations are computed with DuckDB over the parquet store using wallet-hash / block-range bucketing (each aggregation processes a bounded fraction of the data with on-disk spill), so the full ~1.5-billion-row scans run within a fixed memory budget. Every figure is written to a versioned JSON artifact.

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3. Data

3.1 Two eras, near-complete. We use two independent, near-complete on-chain records:

- **v1 archive** (Qin & Yang, 2026, arXiv 2606.04217; CC-BY): the reconciled Polymarket history, ~746 million maker-fill records across ~851k markets and ~2.6 million wallets, Nov 2022 – Apr 2026, with ground-truth aggressor direction, prices, per-fill fee, category, and resolved outcome. This is the *fee-free-to-fee-introduction* era (fees phased in during early 2026).
- **v2 on-chain tape**: our own extraction of the `OrderFilled` / `OrdersMatched` / `FeeCharged` events of the two v2 exchange contracts on Polygon, ~715 million raw / ~697 million deduplicated fills, Apr–Aug 2026 — the *mature-fee* era. Winner outcomes come from a verified on-chain conditional-tokens resolution registry (2,108,903 resolved conditions; token→winner derivation validated for both standard USDC-collateral and negRisk wrapped-collateral markets).

3.2 Coverage and comparability. Both eras cover essentially the whole population, matching the coverage of the concurrent full-population studies (Akey ≈2.4M users; Gómez-Cram ≈1.72M). The differentiator of this work is therefore not data coverage but the fee lens and the actor/ account-type classification applied to that population.

3.3 Known data caveats (carried into every result).

- **Taker-side tail (v1 feed only).** All population results use the complete archive and tape, not the taker-side 4,000-fill API tail, so wallet-level statistics are not subject to that sampling bias.
- **Chunk overlap (v2).** The resumable crawl can duplicate a fill across overlapping files, so all v2 aggregates deduplicate on (tx-hash, event-index).
- **Fee location.** The taker fee lives on `is_taker_order = TRUE` records, not the maker-fill rows; net-of-fee PnL uses the former.
- **Survivorship / outcome-dependence** (§2.1) applies throughout; results are descriptive.
- **Fees vs. eras.** The v1 window is fee-free until the early-2026 rollout (gross ≈ net for its 2022–2025 bulk); v2 carries a ~1.17% effective taker fee, so cross-era PnL comparisons use v2 net of fees.

- **Fee source.** The v1 archive's per-row fee column does not follow the on-chain fee formula, so fee levels are taken from the v2 on-chain fee (which matches the documented `feeRate·p·(1−p)·size`); the archive-era fee is reconstructed from the documented formula and the per-category rate, and the archive's rate field dates the staggered rollout.

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4. State-of-the-art taxonomy

This section maps the recent literature against the analytical dimensions this work touches, dimension by dimension, so the gap the research questions address is visible directly in the table.

4.1 Coverage matrix

Work (year, id)	Population-scale	Who-wins / concentration	Skill test	Maker vs. taker	Fee incidence / behavior	Maker-subsidy capture / adverse selection	On-chain account TYPE	Behavioral-actor taxonomy
Akey et al. (2026, SSRN 6443103)	✔ full	✔ (top-1% ≈76.5%)	mixture model	✔ maker-share = top predictor (continuous)	✗	partial (maker-share→profit, no AS mechanism)	✗	✗ (continuous features, no discrete classes)
Gómez-Cram et al. (2026, SSRN 6617059)	✔ full	✔	✔ sign-randomization (~3% skilled)	✔ MMs separated functionally	✗	partial (MMs+skilled capture >30%)	✗	partial (MM vs skilled, skill-based)
Nechepurenko (2026, arXiv 2605.11640)	✗ (~2.3-day)	✗	✗	maker-side only	✗	✗	✗	attempted, then retracted (fills can't identify MM)
Mitts & Ofir (2026, SSRN 6426778)	large	✗	✗ (informed-trading score)	✗	✗	✗	wallet *provenance* (new wallets), not TYPE	✗
Whelan (working paper)	theory	✗	✗	✔ maker/taker economics	✔ theory only (no on-chain incidence)	partial (theory)	✗	✗
Qin & Yang (2026, arXiv 2606.04217)	✔ (dataset)	✗	✗	✔ (aggressor direction)	✗	✗	✗	✗
Rahman et al. (2025, arXiv 2510.15612)	SoK	—	—	—	lists as open	lists as open	✗	✗
This work	✔ full (both eras)	✔ (replicates)	✔ (reconciles)	✔	✔ incidence (empirical); behavior null (context)	✔ identified markout: makers +, spread capture, no fill-time AS; who captures	✔ EOA/Safe/proxy/custom	✔ MM/HFT/active/retail (heuristic)

(✔ = addressed; partial = touched but not the focus; ✗ = not addressed.)

4.2 What each did and did not do

- The full-population papers (Akey; Gómez-Cram) own who-wins, concentration, skill, and the *maker-advantage direction*; they do not study the fee, the adverse-selection mechanism, the on-chain account type, or a discrete behavioral-actor taxonomy.
- Nechepurenko is the only prior fill-side behavioral attempt and it disclaims its own tiers on identifiability grounds — a direct methodological warning we adopt, and a gap we partly fill by adding profit and the whole population (while conceding the same identification limit for MM/HFT).
- Whelan supplies the theoretical fee framing but no on-chain incidence.
- Mitts & Ofir use wallet *provenance* (age) for insider detection — adjacent to, but not, an account-*type* taxonomy.

4.3 The gap the RQs address

Two columns of the matrix are empty across the entire literature: *fee incidence / behavior* and *maker-subsidy capture / adverse selection*; a third, *on-chain account type*, is empty as a research taxonomy — it exists only as documented product infrastructure, Polymarket's `signature_type`. The first empty column is Sub-RQ 1 (the incidence of the fee). The second and third together are Sub-RQ 2 (who captures the fee-funded subsidy and the on-chain account-type identity of the beneficiary). Conversely, the *who-wins / concentration / skill / maker-vs-taker* columns are filled — and filled specifically for the pre-fee market, since the definitive who-wins study ends on the eve of the fee — so we treat those results as a replication backdrop, cited to Akey and Gómez-Cram, never as our contribution. The lead contribution is therefore the fee experiment and the maker-subsidy mechanism, with the account-type identity as the structural layer within it.

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5. Results

All figures are computed on the full datasets described in §3.

5.1 A regressive fee: the smallest orders bear the highest rates

Reading incidence per taker-order-size decile in the mature (v2) regime, the effective fee declines from ~2.7–3.2% of notional on the smallest orders to ~1.0% on the largest (`fee_behavioral.json`) — a broadly declining gradient, with deciles 1–4 at 2.7–3.2% and decile 10 at 1.0%, not strictly monotone through the middle deciles. This is the fee the documented formula implies, and the v2 tape confirms it to the basis point: the on-chain taker fee equals `feeRate · p · (1−p) · size` with `feeRate ≈ 0.07` (the median of `fee ÷ [p·(1−p)·size]` over v2 taker orders is 0.0700), so fee-over-notional equals `feeRate · (1−p)`, is bounded by the headline rate, and falls as larger orders execute at prices closer to the extremes — the smallest, price-taking orders bear the highest rate. Fewer than 0.2% of v2 taker orders exceed the ceiling (rounding and wrapped-collateral edges). The taker rate varies by category — 7% on crypto, 5% on sports, economics, culture, weather and other, 4% on finance, politics, tech and mentions, and zero on geopolitics — so the gradient is a blend of category rates. Reconstructed from the documented formula and the per-category rate, the v1-era staggered rollout

reproduces the same regressive gradient across both eras. Over the mature-fee window (Apr–Aug 2026) the platform collected ≈$149.5M in taker fees (`v2_platform.json`). The fee is thus real, sharp at introduction, and regressive — it taxes small, price-taking flow proportionally hardest. *(Sub-RQ 1.)*

5.2 Market context: no participation response is identified once pre-trends are removed

A balanced event study — the six market categories present at every month of a [−3, +1] window around each category's own fee-start, compared at constant composition (`fee_rollout_did.json`, `event_study_balanced`) — finds no participation response attributable to the fee. Median taker order size was **already falling before any fee** (pre-fee slope ≈ −$0.51/month); across the window the per-category mean runs 5.68 → 5.96 → 4.42 → 3.85 → 3.61, and the decline at fee onset does not exceed that pre-trend (−$0.57 from month −1 to 0 against a −$0.51 trend) and in fact *decelerates* as the fee ramps up (−$0.24 from 0 to +1). New-entrant inflow was **rising** pre-fee (≈ +4,400 wallets per category per month) and does not fall at onset — per category it runs 32,664 (−1) → 46,542 (0) → 37,744 (+1). On the balanced panel the volume/participation response is therefore **not identified**, consistent with the near-null elasticity expected for a small transaction cost. What is robust is the fee's sharp onset itself — the effective rate rises from zero before fee-start to a positive, climbing level as the schedule ramps. The welfare case for the fee thus rests on its **incidence and distribution** (§5.1, §5.3–§5.4), not on a participation collapse. *(Market context; see §1.2.)*

5.3 The fee is a transfer to high-volume makers, who capture the spread — not by avoiding adverse selection

On the maker side of the archive, 637,314 of 1,214,275 maker wallets (52.5%) are net-negative in realised PnL (`maker_economics.json`) — just over half of all liquidity providers *lose* — and positive maker PnL is hyper-concentrated (Gini ≈ 0.95; the top 1% of winning makers take ~69%). Makers as a class net +$86.0M, the exact mirror of the takers' −$86.1M, and the surplus sits in the top volume decile (~+$1,061 mean, +$128.8M aggregate — more than the entire net maker profit), while the upper-middle (7th–9th) volume deciles carry essentially all the aggregate losses (−$24.4M, −$12.0M, −$9.9M). Realised maker profit is thus non-monotonic in volume and concentrated in the largest makers.

The *mechanism* for that edge is spread capture. An identified signed post-fill markout on the mature (v2) tape — for each maker fill, the token's forward price minus the fill price, signed by the maker's side, across 42M maker fills — is **positive at every horizon from 60 s to one day** (dollar-weighted +0.002 to +0.006 per share): makers systematically earn the post-fill move, capturing the spread rather than being picked off by informed flow. The maker edge is thus not adverse-selection avoidance; across makers the correlation of volume with markout is ≈ 0, and the only trace of adverse selection is a transient effect on the smallest makers at the 60-second horizon that reverts within five minutes. The realised-PnL dispersion among makers is therefore not driven by fill-time picking-off — the execution component is favourable across the maker population — but by position and inventory selection and the eventual outcome.

Combined with §5.1–§5.2 this frames the fee as a regressive transfer whose proceeds, together with the bid-ask spread, flow from price-taking takers to the high-volume makers who capture them. The fee-funded rebate is allocated by Polymarket's published rule in proportion to the fee value each maker's liquidity generates, so it accrues to these same high-volume makers. The *direction* — liquidity provision earns the edge — replicates Akey and Gómez-Cram; the new element is the identified markout showing the maker edge is spread capture, with adverse selection only a negligible, transient, small-maker effect. *(Sub-RQ 2, capture and mechanism.)*

5.4 Who captures, structurally: the on-chain account map and the actors behind the edge

Account type. Resolving on-chain contract types on an activity- and PnL-stratified classified sample (n≈27,111) shows the population is ~97% smart-contract proxy wallets, split by onboarding channel: ~74.5% Gnosis Safe (browser-wallet users, `signature_type` 2), ~21% Polymarket factory proxy (email/"Magic" users, `signature_type` 1), ~3.3% EOA, and a ~1% tail of minor-variant contracts (`account_types.json`). Winners are almost exclusively the two dominant standard proxy types; the minor-variant contracts are low-activity one-offs, and we do not claim they are or are not bots. Mean realised PnL is highest for the email/Magic proxy (~$65k) — but this is driven by a few extreme winners: the median account of every type is a net loser (−$1 to −$12), so account type by itself does *not* separate winners from losers. It is an onboarding-channel/infrastructure axis, not a skill axis.

Behavioral actors. Classifying active wallets by function (`actor_taxonomy.json`, 1.52M wallets):

actor type	n	maker-share	fills/day	24-7 (entropy frac)	mean PnL	win-rate
market_maker	84,120	0.84	16	0.72	+$1,991	47%
hft_bot (near-24/7)	50,441	0.22	109	0.96	+$1,152	29%
active_trader (directional)	274,031	0.13	32	0.63	−$312	21%
retail_casual	1,086,388	0.18	4	0.67	−$89	32%
one_shot	23,961	0.11	13	0.31	−$346	23%

As a class, market-makers (+$167.5M aggregate) and high-frequency near-24/7 automated accounts (+$58.1M) are the net winners — capturing the edge via spread capture, not forecasting — while directional and retail classes lose in aggregate (−$85.4M active, −$96.5M retail). Within *every* class the median account is a small net loser, so the class-level edge sits in each class's right tail (mean ≫ median). This is precisely the population that captures the maker subsidy in §5.3. Two qualifications apply: the economic *direction* — liquidity provision and automation earn the edge — replicates the concurrent papers, and, because public fills do not carry the order/quote lifecycle (Nechepurenko, 2026), these are functional behavioral classes rather than identified market-maker roles. The new-to-the-literature contribution is the on-chain account-type axis and its crossing with function and PnL — most EOAs behave as retail (~62%), with

a minority (~6% of the HFT class, ~1.8× over-represented) automated, so account type maps to *onboarding channel*, not cleanly to bot-vs-human. *(Sub-RQ 2, structural identity.)*

5.5 Supporting / replicated backdrop (cited, not claimed as novel)

For completeness and to situate §5.1–5.4: takers lose −$86.1M in the ~fee-free v1 era; in the fee era they are gross-positive (+$20.0M) but −$129.5M net — the ~1.17% taker fee (≈$149.5M) is the entire loss channel and then some, so the loss mechanism *shifts* from the maker spread (v1) to the fee (v2). Winnings are hyper-concentrated (Gini ≈ 0.95; the top 1% of winning wallets take ~70% of all winnings) — replicating Akey. On skill, a full-population sign-randomization test yields ~4.3% certifiably skilled after Benjamini–Hochberg correction (7.1% uncorrected) — the same order as Gómez-Cram's ~3%, and an upper bound given cross-market outcome correlation; the correlation-immune signal is weak-but-real persistence (split-half Spearman ≈ 0.15, p≈0; period-1 top-quintile → +$1,174 mean in period 2 vs −$134 for the bottom). Skill is expressed as breadth — the skilled tier trades far more markets (mean ~314 vs ~113 neutral) at a much higher market-level win-rate (~85%), not one lucky bet; a stricter position-level test in the informative price band is by contrast null (observed skilled winners 123 < chance 257), reinforcing the upper-bound reading. Coordination is a negative result: funding-graph entity collapse is small (~4.3% of top-20k-winner diversity is sybil), dominated by a single 377-wallet cluster (+$48M) with the remainder singletons — no broad coordinated syndicate, consistent with the project's prior coordination null. These support, but are not, the contribution.

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6. Limitations

- **Participation response.** The event study holds category composition constant and removes the fee-free pre-trend; on that basis no participation or volume response to the fee is identified. The rollout affords only about nine treated categories, too few for cluster-robust inference, so this is an underpowered null, not a demonstrated absence of effect, and the incidence and capture results do not depend on it.
- **Descriptive, outcome-dependent PnL.** All profit-and-loss is realised against the resolved outcome and subject to survivorship; it measures realised performance, not risk-adjusted skill. The dispersion in realised PnL among makers reflects position and inventory selection and the eventual outcome, not the fill-time execution component, which is favourable across the maker population (§5.3).
- **Behavioral classes and account type.** Market-maker and HFT labels are functional classes inferred from public fill activity, which does not carry the order/quote lifecycle (Nechepurenko, 2026), so they are behavioral types, not identified roles. On-chain account type resolves an onboarding and infrastructure channel, not a beneficial owner or human-versus-bot status.

- **Rebate capture is argued from the published rule.** The maker rebate is allocated by Polymarket's published fee-curve rule in proportion to the fee value each maker's liquidity generates, so it accrues to the high-volume makers shown here to capture the spread (§5.3–§5.4). This is an inference from the published rule and the measured concentration of the maker edge, not a recipient-level accounting of the rebate distribution.

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